2026 Spring

Course ID: 1142-1764

Foundations of Computer Science

03-Computer Organization and Cloud Computing

1

HOUR

10:10 – 11:00

Topic

Course Introduction

Cloud Computing

Course Introduction

Welcome to the course — let's get started.

Week 1

50 minutes

Course Info

Introduction to Computer Science

Fundamentals · Organization · Networks · OS · Algorithms · Data Structures · ADT

2 Semesters · 11 Topics · 32 Weeks · 3 Credits each

INSTRUCTOR	DEPARTMENT	LANGUAGE	TOOLS
Prof. H.T. Chu	**International Program in Computer Science**	**English (EMI)**	**AWS Academy**

ASSESSMENT (EACH SEMESTER)

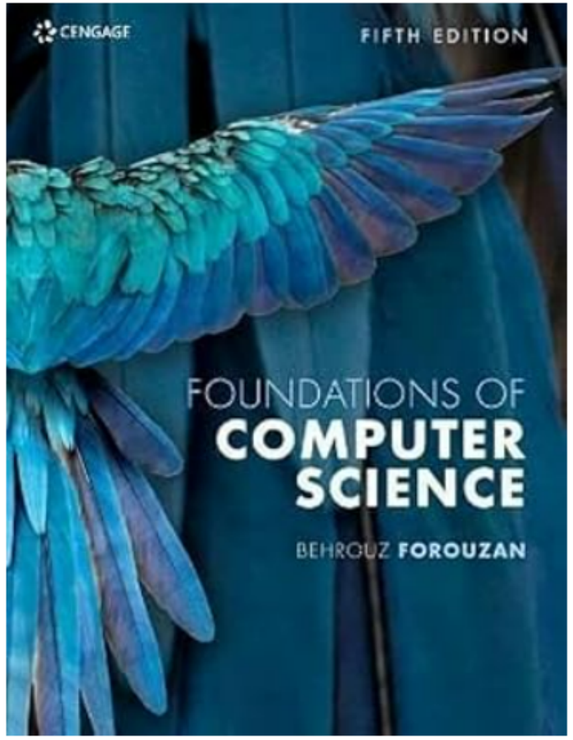
Assignments	Midterm Exam	Project / Labs	Final Exam	Participation
30%	**20%**	**30%**	**20%**	**Bonus 5%**

Providence University · Department of Computer Science and Information Engineering · AY 2025–2026

Thursday, 3-4 (10:10-12:00)

Providence Hall R307

The Textbook



Foundations of Computer Science Paperback

作者 Behrouz Forouzan (Author)

The symposium was held Oct.-Nov. 1989, Research Triangle Park, North Carolina. One hundred papers in theoretical computer science treat dispersers, deterministic amplification, and weak random sources; efficient NC algorithms for set cover with applications to learning and geometry; the inverse of automorphism in polynomial time; and speeding-up linear programming using fast matrix multiplication. Acidic paper; no subject index. Annotation copyright Book News, Inc. Portland, Or.

報告此產品或賣家的問題

語言	尺寸	ISBN-10	ISBN-13
English	18.9 x 24.7 x 2.6 cm	1473787327	978-1473787322

查看所有詳細內容

1 Introduction

Turing Model
Von Neumann Model
Computer Components
History
Social and Ethical Issues
Computer Science as a Discipline
Outline of the course
End-Chapter Materials
Practice Set

2 Number Systems

Introduction
Positional Number Systems
Non-Positional Number Systems
End-Chapter Materials
Practice Set

3 Data Storage

Data Types
Storing Numbers
Storing Text
Storing Audio
Storing Images
Storing Video
End-Chapter Materials
Practice Set

4 Operations on Data

Logic Operations
Shift Operations
Arithmetic Operations
End-Chapter Materials
Practice Set

5 Computer Organization

Introduction
Central Processing Unit
Main Memory
Input/Output Subsystem
Subsystem Interconnection
Program Execution
Different Architectures
A Simple Computer
End-Chapter Materials
Practice Set

6 Computer Networks

Introduction
TCP/IP Protocol Suite
Layers
Internet Applications
The World Wide Web (www)
End-Chapter Materials
Practice Set

7 Operating Systems

Introduction
Evolution
Components
A Survey of Operating Systems
End-Chapter Materials
Practice Set

8 Algorithms

Concept
Three constructs
Algorithm Representation
A More Formal Definition
Basic Algorithms
Subalgorithms
Recursion
End-Chapter Materials
Practice Set

9 Efficiency of Algorithms

Definition
Common Algorithms
Big-O Notation
End-Chapter Materials
Practice Set

10 Programming Languages

Evolution
Translation
Programming Paradigms
Common Concepts
End-Chapter Materials
Practice Set

11 Software Engineering

The Software Life cycle
Analysis Phase
Design Phase
Implementation Phase
Testing Phase
Documentation
End-Chapter Materials
Practice Set

12 Data Structure

Arrays
Records
Linked Lists
End-Chapter Materials
Practice Set

13 Abstract Data Types

Background
Stacks
Queues
General Linear Lists
Trees
Graphs
End-Chapter Materials
Practice Set

This semester

6 Computer Networks

Introduction
TCP/IP Protocol Suite
Layers
Internet Applications
The World Wide Web (www)
End-Chapter Materials
Practice Set

7 Operating Systems

Introduction
Evolution
Components
A Survey of Operating Systems
End-Chapter Materials
Practice Set

8 Algorithms

Concept
Three constructs
Algorithm Representation
A More Formal Definition
Basic Algorithms
Subalgorithms
Recursion
End-Chapter Materials
Practice Set

9 Efficiency of Algorithms

Definition
Common Algorithms
Big-O Notation
End-Chapter Materials

12 Data Structure

Arrays
Records
Linked Lists
End-Chapter Materials
Practice Set

13 Abstract Data Types

Background
Stacks
Queues
General Linear Lists
Trees
Graphs

+ aws academy

Computer Organization

& Cloud Computing

From Transistors to the Cloud

💻 **Computer Org.**

⚙️ **Hardware vs DC**

☁️ **Cloud Services**

50 Slides | Computer Science Fundamentals | Providence University

3	50	100+
Parts	Slides	Concepts

Table of Contents

Course Overview — 3 Sections · 50 Slides

Section 01

Computer Organization

Slides 3–18

- Von Neumann Architecture
- CPU: ALU, Control Unit, Registers
- Memory Hierarchy
- Storage: HDD / SSD / NVMe
- Buses, I/O & Performance Metrics

Section 02

Hardware Comparison

Slides 19–30

- PC vs. Data Center Overview
- CPU: Consumer vs. Server
- Memory: DDR5 vs. RDIMM/HBM
- Storage Arrays & RAID
- Power, Cooling & Reliability

Section 03

Cloud Computing

Slides 31–50

- IaaS / PaaS / SaaS Models
- Virtualization & Containers
- AWS Core Services
- Auto-Scaling & Serverless
- Security, Cost & Future Trends

SECTION 1

Computer Organization

Hardware Fundamentals — How a Computer Really Works

Von Neumann Architecture

04 The Foundation of Modern Computers

1945

Proposed by John von Neumann — established the blueprint for every modern computer

Stored Program

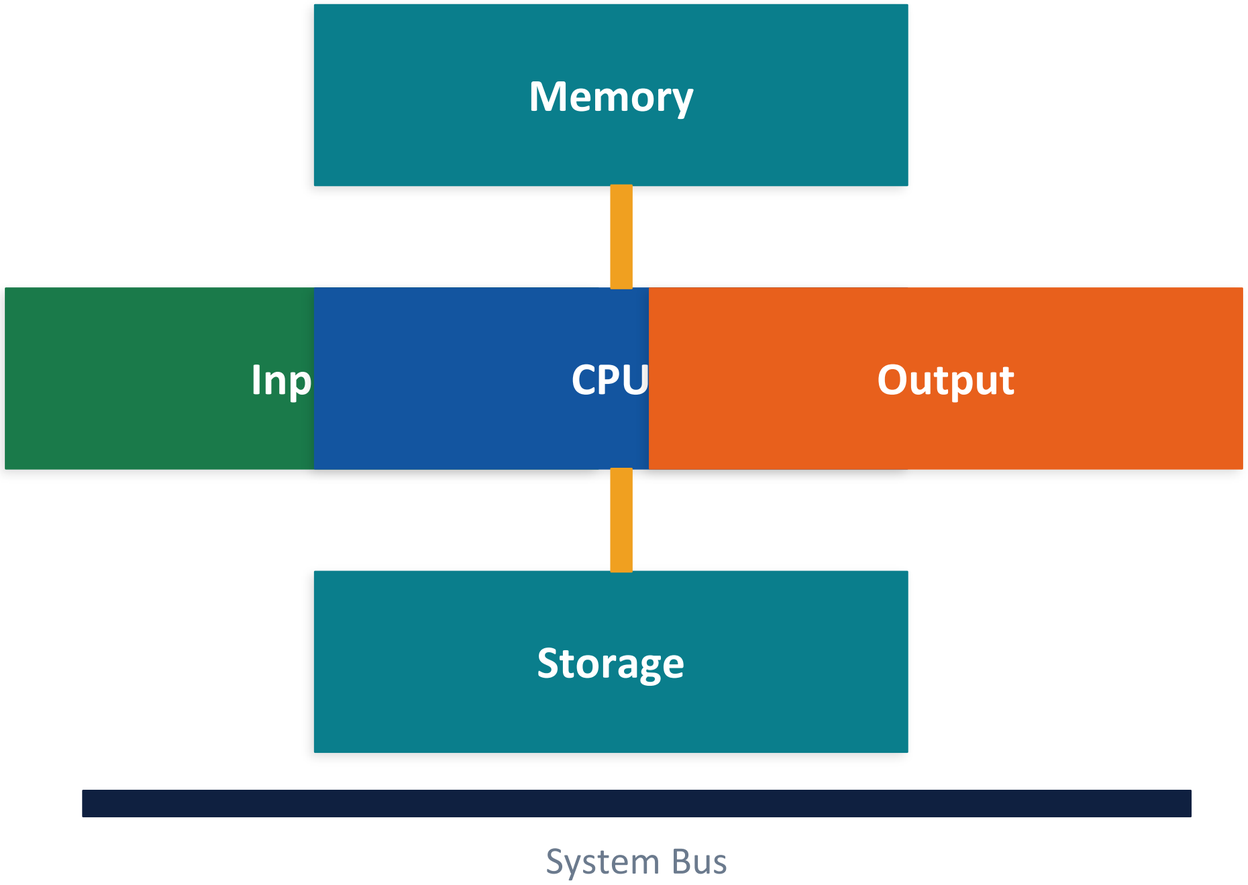
Instructions and data share the same memory; CPU reads and executes sequentially

Four Components

CPU, Memory, I/O Devices, and Buses cooperate to execute programs

Fetch-Decode-Execute

The instruction cycle: Fetch → Decode → Execute → Write-Back repeats endlessly



CPU Architecture & Components

05 Inside the Central Processing Unit

Arithmetic Logic Unit (ALU)

Core computation engine

Performs arithmetic (+, −, ×, ÷) and logical operations (AND, OR, NOT, XOR) on binary data

Control Unit (CU)

Orchestrator of the CPU

Decodes instructions, generates timing signals, and coordinates all CPU sub-components

Register File

Fastest storage on-chip

Tiny, ultra-fast storage (<1 ns); holds operands, addresses, and flags during execution

Cache Memory (L1/L2/L3)

Speed buffer for RAM

Bridges the gap between fast registers and slow RAM; reduces average memory access time

Instruction Pipeline

Parallel execution stages

Overlaps Fetch / Decode / Execute / Write-Back stages to improve throughput (IPC)

Branch Predictor

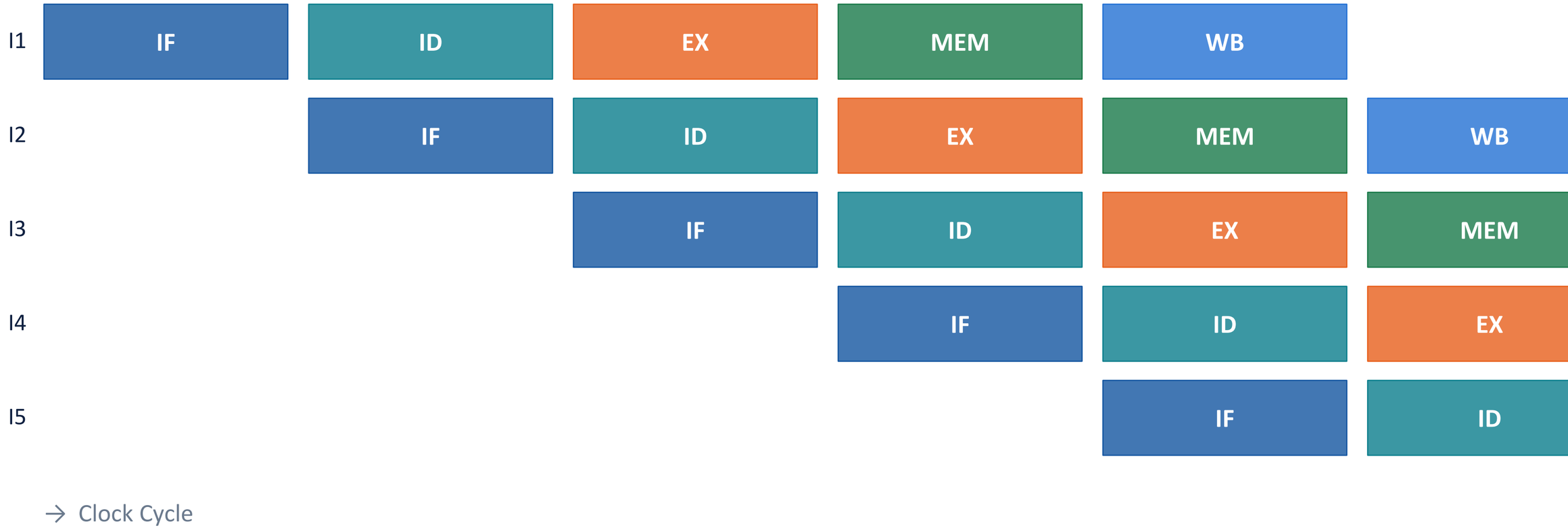
Speculative execution

Guesses conditional branch outcomes to keep the pipeline full and avoid costly stalls

Instruction Cycle & Pipelining

06 How the CPU Executes Instructions in Parallel

Pipeline Diagram — 5 instructions executing concurrently across 5 stages:



CPU Performance Metrics

07 How We Measure Processing Speed

GHz	Core	IPC	TDP
Clock Speed (Cycles / Second)	Core Count (Parallel Threads)	Instructions Per Cycle (Architecture Efficiency)	Thermal Design Power (Watts dissipated)

CPU Performance Formula	`Perf = Clock_Speed × IPC × Core_Count`	Improving any factor boosts overall performance
CPI — Cycles Per Instruction	`CPI = Total_Cycles ÷ Instruction_Count`	Lower CPI means higher execution efficiency
MIPS — Millions of Instructions / sec	`MIPS = Clock_Speed(MHz) ÷ CPI`	Common cross-architecture comparison metric
Amdahl's Law	`Speedup = 1 / [(1−P) + P/N]`	P = parallel fraction, N = processor count

Memory Hierarchy

08 The Speed-vs-Capacity Trade-off

Registers	<1 ns	<1 KB
L1 Cache	1–4 ns	32–64 KB
L2 Cache	4–10 ns	256 KB–1 MB
L3 Cache	10–40 ns	4–32 MB
Main RAM (DRAM)	50–100 ns	8–512 GB
SSD (NVMe)	50–100 µs	256 GB–8 TB
HDD	5–10 ms	500 GB–20 TB

← Faster / Smaller / More Expensive

Slower / Larger / Cheaper →

RAM Types & Specifications

09 Consumer to Enterprise Memory

SRAM — Static RAM

Speed:	**<1 ns**
Capacity:	**KB–MB**
Use Case:	**CPU L1/L2/L3 Cache**

DRAM — Dynamic RAM

Speed:	**~50 ns**
Capacity:	**GB scale**
Use Case:	**System main memory**

DDR5 — Double Data Rate 5

Speed:	**51.2 GB/s**
Capacity:	**8–64 GB**
Use Case:	**Desktop / Laptop**

LPDDR5 — Low-Power DDR5

Speed:	**51.2 GB/s**
Capacity:	**4–32 GB**
Use Case:	**Smartphones / Laptops**

RDIMM — Registered DIMM (ECC)

Speed:	**57.6 GB/s**
Capacity:	**16–256 GB**
Use Case:	**Server standard**

HBM3 — High Bandwidth Memory 3

Speed:	**819 GB/s**
Capacity:	**16–96 GB**
Use Case:	**GPU / AI Accelerators**

Storage Device Comparison

10 HDD vs SATA SSD vs NVMe vs Optane

Specification	HDD	SATA SSD	NVMe SSD	Optane / PMem
Interface	SATA 6Gb/s	SATA 6Gb/s	PCIe 4.0 x4	PCIe / DDR
Read Speed	150 MB/s	550 MB/s	7,000 MB/s	7,000+ MB/s
Write Speed	120 MB/s	500 MB/s	6,000 MB/s	3,500 MB/s
Latency	5–10 ms	50–100 µs	20–100 µs	<10 µs
$/GB	~$0.02	~$0.08	~$0.12	~$3.00
Capacity	500 GB–20 TB	128 GB–4 TB	256 GB–8 TB	16–512 GB
Best For	Bulk storage	Balanced use	OS / workloads	Persistent memory

Motherboard & Chipset

11 The Hub That Connects Everything

CPU Socket

Physically houses the CPU
Intel LGA 1851 / AMD AM5 (2024)

RAM Slots

DDR5 DIMM slots (2–8 slots)
Dual-channel / quad-channel configs

PCIe Slots

Expansion for GPU, NVMe, NIC
PCIe 5.0 provides 128 GB/s per x16

Chipset (PCH)

Mediates CPU ↔ peripherals
Intel Z790 / AMD X670E

BIOS / UEFI

Firmware stored in Flash ROM
Power-On Self Test (POST) + boot

VRM

Voltage Regulator Module
Delivers clean, stable CPU power

SATA Ports

Connect HDD / SATA SSDs
6 Gbps transfer rate

USB / I/O Panel

Rear panel connectors
USB4, Thunderbolt 4, 2.5G LAN

Buses & I/O Interfaces

Data Bus

Carries actual data; width (64-bit) determines how much data per transfer

Address Bus

Specifies memory locations; width determines maximum addressable space

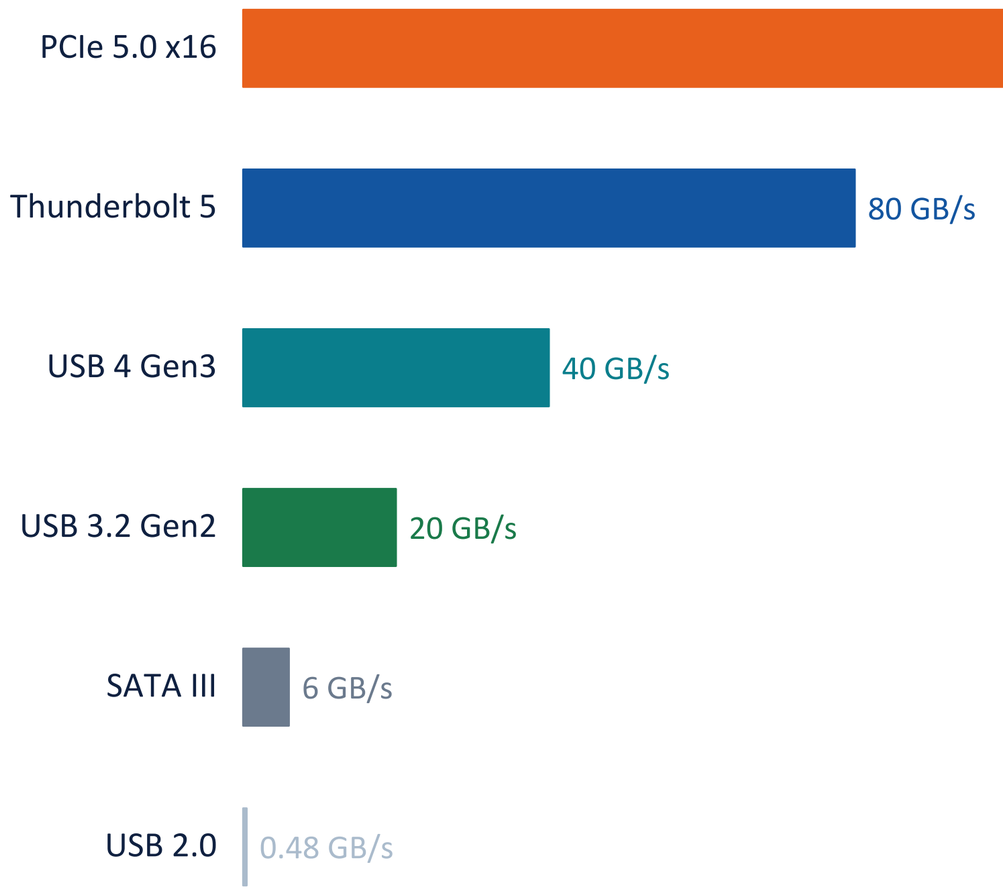
Control Bus

Carries read/write signals, interrupts, and clock signals between units

PCIe Bus

High-speed serial link; Gen 5 x16 delivers 128 GB/s for GPUs & NVMe

Interface Bandwidth Comparison



Number Systems & Data Representation

13 How Computers Store and Process Information

Base-2	Base-8	Base-10	Base-16
Binary	**Octal**	**Decimal**	**Hexadecimal**
Digits: 0, 1	Digits: 0–7	Digits: 0–9	Digits: 0–9, A–F
$1010_2 = 10_{10}$	$12_8 = 10_{10}$	$10_{10} = 10_{10}$	$0A_{16} = 10_{10}$

Data Units

1 bit Smallest unit: 0 or 1	**1 Byte = 8 bits** One text character	**1 KB = 1,024 B** Small text file	**1 MB = 1,024 KB** Photo / song
1 GB = 1,024 MB Video / OS install	**1 TB = 1,024 GB** Large database	**1 PB = 1,024 TB** Data center scale	

Role of the Operating System

14 The Software Layer Between Hardware and Applications

Process Management

Scheduling, creation/termination, inter-process communication (IPC), deadlock handling

Memory Management

Virtual memory, page replacement, memory protection, heap/stack management

File System

FAT32, NTFS, ext4 — directory trees, file permissions, journaling

Device Drivers

Hardware abstraction layer, driver interfaces, interrupt handling (IRQ), DMA

Security & Protection

User privilege rings, kernel/user mode isolation, access control lists (ACL)

Network Stack

TCP/IP implementation, Socket API, firewall, virtual network interfaces

GPU & Hardware Accelerators

15 Specialized Compute Beyond the CPU

CPU vs GPU Architecture

- CPU: few powerful cores (8–64), optimized for serial tasks
- GPU: thousands of small cores (CUDA), optimized for parallel math
- CPU = low latency; GPU = high throughput

GPU Use Cases

- 3D rendering: games, CAD, video editing
- AI/ML training: highly parallel matrix ops
- Scientific computing: fluid dynamics, protein folding
- NVIDIA H100: 4 PFLOPS FP8 for LLM training

NVIDIA GPU Generations

- Ampere (A100): 312 TFLOPS — AI training mainstay
- Hopper (H100): Transformer Engine, 4 PFLOPS
- Blackwell (B200): next-gen AI supercomputing arch
- DGX H100: 8× H100, NVLink 3.6 TB/s total

Other Accelerators

- FPGA: programmable logic, ultra-low latency
- TPU: Google's custom AI tensor processor
- NPU: on-device AI inference (smartphones)
- DPU: data center network offload processor

Input / Output Devices

Input Devices

Keyboard

Scan codes → USB HID protocol → OS keyboard driver

Mouse

Optical sensor detects movement; DPI determines precision

Camera

CMOS sensor + ISP image signal processor

Microphone

Sound wave → ADC → digital audio stream

Sensors

Temp / accelerometer / gyroscope → IoT applications

Output Devices

Display

GPU HDMI/DP output → LVDS/eDP panel driver

Printer

GDI/PostScript → laser or inkjet printing

Speaker

DAC → power amplifier → speaker cone

Projector

DLP/LCD optics, brightness in ANSI lumens

LED Wall

Dot-matrix controller, refresh rate ≥ 1920 Hz

Part 1 Summary — Computer Organization

17 Key Takeaways Before Moving On

01

Von Neumann Model

CPU + Memory + I/O + Bus; programs and data share a single address space

02

CPU Internals

ALU executes operations; CU orchestrates timing; Registers hold live data

03

Memory Hierarchy

Registers → L1 → L2 → L3 → RAM → SSD → HDD: speed and capacity trade-off

04

Pipelining

Fetch/Decode/Execute overlap boosts throughput; branch prediction reduces stalls

05

Buses & I/O

PCIe 5.0 leads internal bandwidth; USB4/Thunderbolt for external connectivity

06

Performance Factors

Clock × IPC × Cores; Amdahl's Law limits the benefit of parallelism

SECTION 2

[illegible]

Personal Computer vs. Data Center — Deep Dive

PC vs. Data Center — Overview

19 How Hardware Scales from Desktop to Hyperscaler

Specification	Home PC	Workstation	1U Server	Data Center Rack
CPU Type	Consumer (i9/R9)	Xeon W/Threadripper	Xeon/EPYC 1P	Xeon/EPYC 2P–4P
CPU Cores	8–24 cores	32–64 cores	32–96 cores	64–192+ (×2 sockets)
Memory	16–64 GB DDR5	128–512 GB RDIMM	512 GB–4 TB	Hundreds of TB
Storage	NVMe 1–4 TB	NVMe RAID	SAS/NVMe RAID	Distributed PB-scale
Networking	1G/2.5G Ethernet	10G/25G SFP+	25G/100G	400G/800G InfiniBand
Power Supply	500W–1000W single	1500W Redundant	2000W Dual-redundant	N+1/2N Rack PDU
Cooling	Air + fans	Forced airflow	Front-to-rear airflow	Liquid / immersion
Reliability	No redundancy	Partial redundancy	ECC RAM, IPMI	99.999% five-nines

CPU: Desktop vs. Server

20 Consumer Cores vs. Enterprise Compute

Specification	Desktop / Workstation CPU	Server CPU
Representative	Intel i9-14900K / AMD Ryzen 9 7950X	Intel Xeon Platinum 8592+ / AMD EPYC 9965X
Core Count	8–24 P+E Cores	64–192 Cores
L3 Cache	32–64 MB	128–1,152 MB (3D V-Cache)
Memory Channels	2-channel DDR5	8–12 channel RDIMM ECC
TDP	65–253 W	350–500 W
PCIe Lanes	16–24 lanes	128+ lanes
Multi-socket	Not supported (1P only)	2P / 4P / 8P configurations
ECC Memory	Usually not supported	Mandatory ECC / LRDIMM
IPMI / BMC	None	Out-of-band remote management
Lifespan	2–5 yr consumer product	5–10 yr 24/7 operation

Memory: Consumer vs. Enterprise

DDR5-6400 (Consumer)	
Capacity:	**8–64 GB**
Bandwidth:	**51.2 GB/s**
Error Correct.:	**No ECC**
Slots:	**2–4**
Use Case:	**Desktop / Laptop**

RDIMM DDR5 (Registered)	
Capacity:	**16–256 GB**
Bandwidth:	**57.6 GB/s**
Error Correct.:	**Mandatory ECC**
Slots:	**8–16**
Use Case:	**1P/2P Server**

LRDIMM DDR5 (Load-Red.)	
Capacity:	**32–256 GB/stick**
Bandwidth:	**57.6 GB/s**
Error Correct.:	**ECC+Parity**
Slots:	**16–32**
Use Case:	**High-density server**

HBM3 (High Bandwidth)	
Capacity:	**16–96 GB**
Bandwidth:	**819 GB/s**
Error Correct.:	**On-chip ECC**
Slots:	**on-package**
Use Case:	**GPU / AI accelerators**

CXL Memory (New Proto.)	
Capacity:	**Up to TB**
Bandwidth:	**PCIe 5.0 BW**
Error Correct.:	**ECC**
Slots:	**CXL slot**
Use Case:	**Memory pool expansion**

Optane PMem (Persistent)	
Capacity:	**128–512 GB/stick**
Bandwidth:	**6 GB/s**
Error Correct.:	**ECC**
Slots:	**DIMM-compat.**
Use Case:	**Database acceleration**

Storage: Consumer vs. Enterprise

22 NVMe to Petabyte-Scale Arrays

Item	Consumer NVMe	Enterprise NVMe	Enterprise SAS SSD	Storage Array
Model Example	Samsung 990 Pro	Samsung PM9A3	Seagate Nytro	NetApp AFF A900
Read Speed	7,450 MB/s	7,000 MB/s	2,100 MB/s	8+ GB/s aggregate
IOPS	1,000K	1,500K	800K	50M+ (clustered)
Endurance (TBW)	600 TBW (1 TB)	3,600 TBW	36,000 TBW	Design for perpetual replace
MTBF	1.5M hours	2.5M hours	2.5M hours	N+1 / RAID redundancy
Interface	PCIe 4.0 M.2	PCIe 4.0/5.0 U.2	SAS-3 12Gb/s	FC / iSCSI / NVMe-oF
Best For	Desktop / laptop	AI training nodes	Database servers	Enterprise shared storage

Networking: Home vs. Data Center

23 From 1 Gbps to 800 Gbps

Home / SMB Network

- 1G/2.5G/10G consumer NIC
- Wi-Fi 6E / Wi-Fi 7 wireless
- Home router: 100–300 Mbps WAN
- Unmanaged or simple VLAN switch
- Typical latency: 1–10 ms WAN

Enterprise Edge Network

- 10G/25G SFP+ NICs
- Cisco Catalyst / Aruba managed switches
- SD-WAN, MPLS connectivity
- Firewall + IPS/IDS inline
- Latency: <1 ms LAN

Data Center TOR

- 100G/400G QSFP56 NICs
- Top-of-Rack switch per rack
- Leaf-Spine Clos topology
- VXLAN overlay networking
- Latency: <10 µs

HPC / AI Cluster

- InfiniBand HDR 200G / NDR 400G
- NVIDIA NVLink 3.6 TB/s
- RoCEv2 RDMA: <1 µs latency
- All-to-All collective communication
- GPUDirect RDMA for zero-copy transfers

Power, Cooling & Form Factors

Desktop (ATX)

Power: 500W–1,000W

Cooling: Tower air cooling

1U Rack Server

Power: 750W–1,600W

Cooling: High-RPM fans

2U Rack Server

Power: 1,200W–3,000W

Cooling: Forced front-to-rear

4U GPU Server

Power: 3,000W–10 kW

Cooling: Liquid cooling option

Blade Server Chassis

Power: Central shared PSU

Cooling: Chassis-level cooling

Data Center Cooling Methods

Air Cooling (CRAC/CRAH) — **PUE: 1.4–2.0**

Hot/cold aisle containment; most common

Direct Liquid Cooling — **PUE: 1.15–1.3**

Cold plate on CPU/GPU; rear-door heat exchangers

Immersion Cooling — **PUE: 1.02–1.1**

Servers submerged in dielectric fluid; ultra-efficient

Free Cooling / Economizer — **PUE: 1.1–1.3**

Uses outdoor air when ambient temp is low enough

PUE = Total Facility Power ÷ IT Equipment Power. Target: 1.0. Google 2023 PUE = 1.10

Redundancy & Reliability

25 Designing for Five-Nines Availability

Uptime Institute Data Center Tier Standards

	Tier I	Tier II	Tier III	Tier IV
Uptime	**99.671%**	**99.741%**	**99.982%**	**99.995%**
Downtime	**28.8 h/yr**	**22 h/yr**	**1.6 h/yr**	**26 min/yr**
Redundancy	**No redundancy**	**Partial N+1**	**Concurrently maintainable**	**Fault tolerant 2N**

RAID Storage Redundancy Strategies

RAID 0	RAID 1	RAID 5	RAID 6	RAID 10
Striping No redundancy	Mirroring 100% redundancy	Distributed parity N−1 efficiency	Dual parity N−2 efficiency	Mirror+Stripe 50% efficiency
Max performance, zero fault tolerance	Full backup; capacity halved	Tolerates 1 drive failure	Tolerates 2 drive failures	High performance + high reliability

Data Center Scale & Statistics

26 The Numbers Behind Global Infrastructure

10M+
Active servers in AWS alone

3.5 ZB
Global data volume (2025 estimate)

1–2%
Global electricity consumed by DCs

$622B
Global cloud market size (2023)

Company	Data Centers	Est. Servers	Power Draw	PUE
Google	35+	~1M	~5 GW	1.10
Microsoft Azure	200+	3–4M	~5 GW	1.18
Amazon AWS	100+	~3M	~8 GW	1.15
Meta	21+	~2M	~2 GW	1.09
Alibaba Cloud	60+	~1M	~2 GW	1.30

Part 2 Summary — Hardware Comparison

27 Key Differences Between PC and Data Center

01

CPU Differences

Server CPUs: more cores, more channels, ECC, dual-socket — desktop CPUs: high single-core frequency

02

Memory Specifications

RDIMM/LRDIMM mandates ECC error correction to protect data integrity in 24/7 operations

03

Storage Arrays

Enterprise SSDs offer higher TBW endurance; SAS/RAID arrays provide redundancy and high IOPS

04

Network Hardware

Data centers use 100G/400G; InfiniBand provides <1 µs ultra-low latency for HPC clusters

05

Cooling & Power

PUE measures energy efficiency; liquid / immersion cooling achieves PUE as low as 1.02

06

Reliability Design

Tier IV = five-nines uptime; RAID + hot spare + UPS + generators provide multi-layer protection

SECTION 3

Service Models, Virtualization & Major Providers

Service Models, Virtualization & Major Providers

What is Cloud Computing?

29 On-Demand Resources Over the Internet

Cloud computing delivers computing resources — servers, storage, databases, networking, and software — over the internet on a pay-as-you-go basis, eliminating the need to own physical infrastructure.

NIST Five Essential Cloud Characteristics

On-Demand Self-Service

Users provision compute resources automatically without human interaction

Broad Network Access

Available over the network via standard mechanisms from any device

Resource Pooling

Multi-tenant; resources dynamically assigned based on demand

Rapid Elasticity

Resources scale up/down quickly — appear unlimited to the consumer

Measured Service

Usage is monitored, controlled, and reported — pay only for what you use

Cloud Service Models

30 IaaS / PaaS / SaaS

IaaS — Infrastructure as a Service

☁ Cloud manages: Physical servers, Network, Virtualization

👤 You manage: OS, Runtime, App, Data

Examples:

- AWS EC2
- Azure Virtual Machines
- GCP Compute Engine
- Linode / DigitalOcean

✓ Maximum flexibility — full control over the environment

PaaS — Platform as a Service

☁ Cloud manages: Infrastructure + OS + Runtime

👤 You manage: Application + Data only

Examples:

- AWS Elastic Beanstalk
- Google App Engine
- Heroku
- Azure App Service

✓ Accelerates development — no server management needed

SaaS — Software as a Service

☁ Cloud manages: Everything — infra, runtime, app

👤 You manage: Your data, settings, and usage

Examples:

- Google Workspace
- Microsoft 365
- Salesforce
- Dropbox / Zoom

✓ Ready to use — zero operations overhead

Cloud Deployment Models

31 Public / Private / Hybrid / Multi-Cloud

🌐 Public Cloud

Owned and operated by a CSP; shared infrastructure over the internet; pay-as-you-go

✓ Lowest cost, highest elasticity, rapid provisioning, global reach

✗ Data sovereignty concerns, compliance restrictions

🏢 Private Cloud

Dedicated to one organization; on-premises or provider-hosted; fully isolated environment

✓ Highest security, full control, meets financial/medical compliance

✗ High CapEx, requires expert team, limited elasticity

🔀 Hybrid Cloud

Combines public + private cloud via secure links; sensitive data stays on-prem, bursts to public

✓ Best balance of flexibility and security; enables gradual migration

✗ Complex architecture, requires cloud connectivity management

☁️ Multi-Cloud

Uses multiple CSPs simultaneously; selects the best service per workload or region

✓ No vendor lock-in, geographic redundancy, leverage each CSP's strengths

✗ High management complexity, cross-cloud egress costs

Virtualization Technology

Type 1 — Bare-Metal Hypervisor

- Runs directly on hardware — no host OS
- Highest performance and security
- Enterprise & cloud production standard
- Examples: VMware ESXi, KVM, Hyper-V, Xen

Type 2 — Hosted Hypervisor

- Runs on top of a host OS
- Slight performance overhead from host OS
- Ideal for development & testing
- Examples: VirtualBox, VMware Workstation

Virtualization Benefits

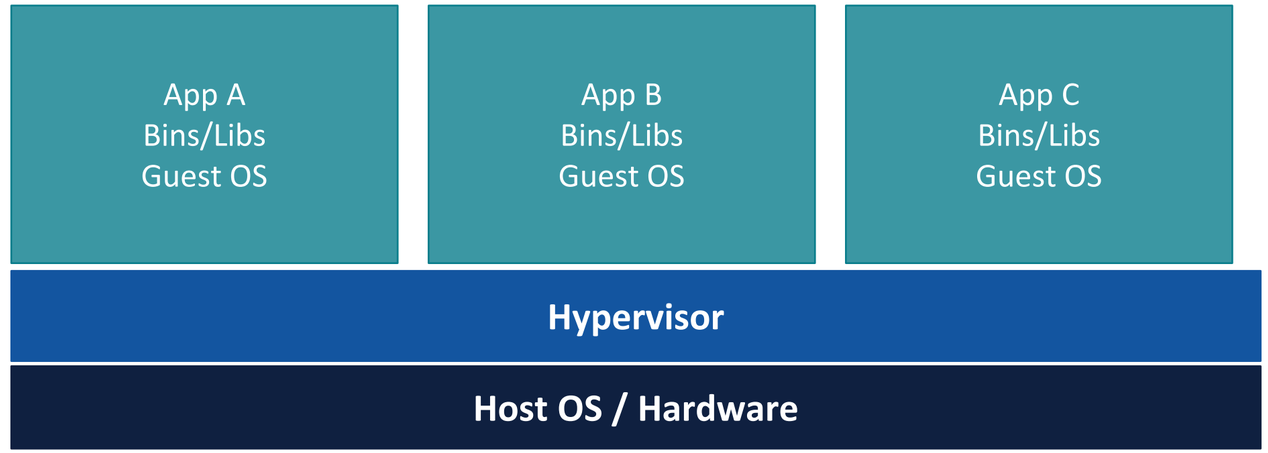
- Server consolidation ratio: 10:1 to 20:1
- Rapid provisioning (minutes vs. weeks)
- Snapshots & live migration (vMotion)
- Disaster recovery: VM-level backup & restore

VM Lifecycle

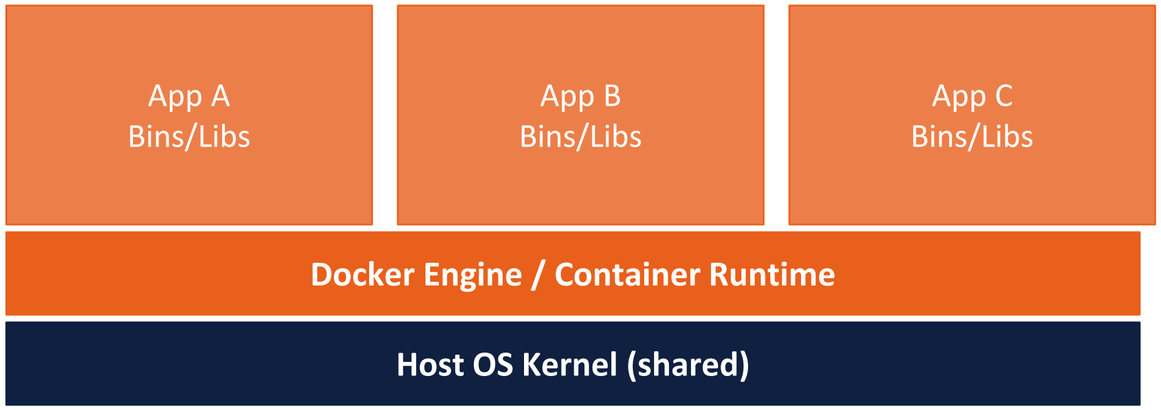
- VM 1: App + Guest OS (isolated)
- VM 2: App + Guest OS (isolated)
- Hypervisor: resource allocation & isolation
- Physical Hardware: CPU, RAM, Storage, NIC

Containers vs. Virtual Machines

Architecture Diagram



Virtual Machines



Containers

Metric	Virtual Machines	Containers	Difference
Start Time	Minutes	Seconds	~100× faster
Image Size	GB-scale	MB-scale	~10× smaller
Isolation	Hardware-level	OS process-level	VMs stronger
Performance	5–10% overhead	<2% overhead	Containers faster
Best Use Case	Legacy apps, multi-OS	Microservices, CI/CD	Complementary

Kubernetes — Container Orchestration

Kubernetes (K8s) is an open-source system for automating deployment, scaling, and management of containerized applications. Originally developed by Google, open-sourced in 2014.

Pod	Deployment	Service	Ingress
Smallest deployable unit; one or more containers sharing network/storage	Manages Pod replicas, rolling updates, and rollback strategies	Provides stable DNS + load balancer for a set of Pods	HTTP/HTTPS routing rules managing external traffic into the cluster

ConfigMap	PersistentVolume	Namespace	Helm Chart
Decouples configuration from code; injected as env vars or files	Provides durable storage that outlives individual Pod lifecycles	Logical cluster partitioning for multi-tenant isolation	Kubernetes package manager — bundles complex applications for easy deploy

Major Cloud Providers Compared

35 AWS vs Azure vs GCP vs Alibaba

	Amazon AWS	Microsoft Azure	Google Cloud	Alibaba Cloud
Market Share:	32%	23%	12%	4%
Founded:	2006	2010	2008	2009
Regions:	33 regions / 105 AZs	60+ regions	40 regions / 121 AZs	28 regions / 88 AZs
Strength:	Largest ecosystem, 200+ services	Enterprise integration, GitHub, Office 365	AI/ML, BigQuery, Kubernetes expertise	Asia-Pacific leader, e-commerce scale
Compute:	EC2	Azure VM	Compute Engine	ECS
Storage:	S3	Blob Storage	Cloud Storage	OSS
Database:	RDS / DynamoDB	Azure SQL / Cosmos DB	BigQuery / Spanner	PolarDB / ApsaraDB
AI/ML:	SageMaker / Bedrock	Azure OpenAI / Copilot	Vertex AI / Gemini	PAI / Tongyi

AWS Core Services Overview

36 The Building Blocks of AWS

Compute

- EC2 — Virtual machines (IaaS)
- Lambda — Serverless functions
- ECS/EKS — Container services
- Fargate — Serverless containers
- Elastic Beanstalk — PaaS

Storage

- S3 — Object storage
- EBS — Block storage (for EC2)
- EFS — Network file system (NFS)
- S3 Glacier — Cold archival storage
- Storage Gateway — Hybrid bridge

Databases

- RDS — Managed SQL (MySQL/Postgres)
- DynamoDB — Serverless NoSQL
- Aurora — High-perf MySQL/Postgres
- ElastiCache — Redis/Memcached
- Redshift — Data warehouse

Networking

- VPC — Virtual Private Cloud
- Route 53 — DNS & traffic routing
- CloudFront — Global CDN (450+ PoPs)
- ALB/NLB — Load balancers
- Direct Connect — Dedicated line

Security

- IAM — Identity & Access Management
- WAF — Web Application Firewall
- Shield — DDoS protection
- KMS — Key Management Service
- GuardDuty — Threat detection

AI / ML

- SageMaker — ML platform
- Bedrock — Foundation model API
- Rekognition — Image/video analysis
- Polly — Text-to-speech
- Lex — Conversational AI (chatbots)

Auto Scaling & Load Balancing

Elastic Scaling Flow

Traffic Spike → CloudWatch Monitoring → Auto Scaling Policy Triggered → New EC2 Instances Launch → Load Balancer Distributes Traffic

Horizontal Scaling (Scale Out)

Add more server instances to distribute load
AWS ASG sets min/max/desired capacity
Triggers on CPU, Memory, Request Count metrics

Vertical Scaling (Scale Up)

Upgrade single server spec (CPU/RAM)
Requires restart; lower elasticity
Suitable for stateful services like databases

Load Balancer Types

ALB (Layer 7): HTTP/HTTPS path-based routing
NLB (Layer 4): Ultra-low latency TCP/UDP
GLB: Cross-VPC security appliance integration

Scaling Policies

Target Tracking: maintain CPU at 70%
Step Scaling: increment by trigger thresholds
Predictive: ML forecasts traffic, scales in advance

Serverless Computing

38 Code Without Managing Servers

Serverless ≠ no servers — the cloud provider fully manages the infrastructure. You upload code; the platform handles provisioning, scaling, and billing per invocation.

AWS Lambda Features

- Triggers: API Gateway, S3, DynamoDB Streams
- Max 15 min execution per invocation
- Memory: 128 MB – 10 GB (controls CPU too)
- Free tier: 1 million requests/month
- Supports Python, Node.js, Java, Go, .NET

Common Use Cases

- Event-driven REST / GraphQL APIs
- Image / video thumbnail generation
- Scheduled batch jobs (Cron)
- IoT sensor data ingestion
- Chatbot & voice assistant backends

Serverless Architecture

- Frontend: S3 + CloudFront (static hosting)
- API: API Gateway + Lambda functions
- Database: DynamoDB (serverless NoSQL)
- Auth: Cognito user pools
- Monitoring: CloudWatch Logs + X-Ray

Limitations & Challenges

- Cold start latency (first-request delay)
- No persistent connections / in-memory state
- Complex distributed debugging
- Vendor lock-in risk
- Cost modeling requires careful analysis

Cloud Storage Services

39 Object, Block, File, and Archive

Object Storage

AWS: S3 **GCP: Cloud Storage** **Azure: Blob Storage**

Unstructured data (photos, videos, backups). 11 nines durability. REST API access.

Use: Static website files, data lake, ML datasets

Block Storage

AWS: EBS **GCP: Persistent Disk** **Azure: Managed Disk**

Acts like a hard drive; attached to VMs. Configurable IOPS and throughput.

Use: Databases, OS root volumes, high-IOPS workloads

File Storage

AWS: EFS **GCP: Filestore** **Azure: Azure Files**

NFS / SMB protocol; multiple VMs share a single file system simultaneously.

Use: Shared config files, CMS media, legacy lift-and-shift

Archive Storage

AWS: S3 Glacier **GCP: Archive** **Azure: Blob Archive**

Lowest cost tier; retrieval takes minutes to hours. For rarely accessed cold data.

Use: Compliance backups, historical logs, regulatory archives

Cloud Networking & VPC

40 Virtual Private Cloud Architecture

VPC (Virtual Private Cloud)

Logically isolated network in the cloud; define your own IP ranges (CIDR), subnets, route tables, and internet gateways

Public & Private Subnets

Public subnet: accessible from internet (requires IGW)
Private subnet: internal only; uses NAT Gateway for outbound access

Security Groups & NACLs

Security Group: stateful, allow rules per instance
NACL: stateless, subnet-level allow/deny rules

VPC Peering & Transit Gateway

Peering: direct VPC-to-VPC connection
Transit Gateway: hub-and-spoke for multi-VPC / multi-account routing

AWS Direct Connect

Dedicated private line from your data center to AWS; more stable and lower latency than VPN; 1G–100G options

CloudFront CDN

450+ global PoPs cache static content; dramatically reduces latency and offloads origin servers

Cloud Security

41 Shared Responsibility & Best Practices

Shared Responsibility Model

☁ CSP Responsible	👤 You Responsible
Physical security (data center, servers)	Data encryption (at rest & in transit)
Hardware infrastructure	IAM user access controls
Hypervisor / virtualization layer	OS patching (IaaS responsibility)
Network infrastructure	Network config (Security Groups / NACLs)
Managed service internals	Application-level security

🔒 Least Privilege

Grant only the minimum IAM permissions needed; rotate access keys regularly

🔒 MFA Everywhere

Enable MFA on root account; enforce MFA policy for all IAM users

🔒 Encrypt All Data

S3 SSE, EBS encryption, TLS in transit; manage keys with KMS

🔒 Network Isolation

Place databases in private subnets; open only required ports in SGs

🔒 Audit Logging

CloudTrail records every API call; enable across all regions

🔒 Continuous Detection

GuardDuty detects threats 24/7; Security Hub aggregates findings

Cloud Economics & Pricing Models

42 CapEx vs OpEx and How to Save

CapEx — Capital Expenditure (On-Premises)	OpEx — Operating Expenditure (Cloud)
⚠ Large upfront hardware purchase cost	✓ Pay-as-you-go — zero upfront capital cost
⚠ Fixed depreciation over 5–7 years	✓ Predictable billing, elastic with usage
⚠ Over-provisioned or under-provisioned	✓ Right-sized resources, auto-adjustable
⚠ Requires dedicated IT operations team	✓ No hardware maintenance overhead
⚠ Long hardware refresh cycles	✓ 30–50% TCO reduction typical

AWS Pricing Models

On-Demand	Reserved 1–3 Year	Spot Instance	Savings Plans
Pay by hour/second No commitment	Commit to usage Steady workloads	Spare capacity Can be interrupted	Commit $/hr spend Flexible service use
0%	Up to 72%	Up to 90%	Up to 66%

DevOps & CI/CD Pipeline

43 From Code Commit to Production in Minutes

Plan	Code	Build	Test
Jira / GitHub Issues	VS Code / Git / GitHub	Maven / Gradle / npm	JUnit / Selenium / k6

Deploy	Operate	Monitor	Feedback
Terraform / Helm / CDK	K8s / ECS / Lambda	CloudWatch / Datadog	Post-Mortem → Plan

GitHub Actions CI/CD Example

```
on: push: branches: [main]  →  jobs: test → build → deploy_to_aws_ecs
Each git push triggers: 1. Unit tests run  2. Docker image built & pushed to ECR  3. ECS service rolling update
```

Edge Computing & IoT

44 Moving Compute Closer to the Data Source

Edge computing shifts processing from centralized data centers to nodes near the data source, dramatically reducing latency and bandwidth consumption.

Device Layer | <1 ms

Sensors, cameras, PLCs, MCUs — on-device inference — Arduino / Raspberry Pi / Coral TPU

Edge Node | 1–10 ms

Factory gateways, retail POS, cellular MEC servers — NVIDIA Jetson / AWS Snowball Edge

Fog / Regional | 10–50 ms

City-scale distributed nodes, 5G MEC — Azure Stack Edge / AWS Outposts

Public Cloud | >100 ms

Global analytics, model training, long-term storage — AWS / Azure / GCP

AWS IoT Stack: IoT Core → Greengrass (edge runtime) → SiteWise (industrial) → IoT Analytics → S3/Redshift

AI & Machine Learning in the Cloud

45 From Data to Deployed Model

Data Collection	Data Processing	Feature Engineering	Model Training	Evaluation	Deploy & Infer
S3 / Kinesis / Glue	EMR / Athena / Pandas	SageMaker Feature Store	SageMaker / GPU Cluster	SageMaker Experiments	SageMaker Endpoint / Lambda

AWS Bedrock

Access Anthropic Claude, Amazon Titan, Meta Llama via API; build GenAI apps without managing ML infra

SageMaker Studio

Unified ML IDE: Jupyter notebooks, AutoML, MLOps pipeline management, model registry

Google Vertex AI

Gemini Pro integration, AutoML, BigQuery ML — strong data-to-AI workflow for analytics-heavy teams

Azure OpenAI

Enterprise-grade GPT-4 / DALL-E / Whisper API with private deployment — data never shared with OpenAI

GPU Clusters

A100/H100 clusters with NVLink interconnect; ideal for training LLaMA 70B, GPT-4 scale models

MLOps Toolchain

MLflow experiments, DVC versioning, ArgoCD deploy, Prometheus monitoring — full production ML lifecycle

Future Trends in Cloud & Computing

46 What's Coming Next

⚛️ Quantum Cloud Computing

IBM Quantum & AWS Braket provide quantum circuit simulation; exponential speedup for optimization; practical use expected ~2030

💡 Silicon Photonics

Optical interconnects replace copper; ultra-low latency; Intel Bare Metal GPU / AMD Instinct next-gen architectures

🧠 Neuromorphic Chips

Intel Loihi & In-Memory Computing break the memory wall bottleneck; brain-inspired sparse computation

🌱 Sustainable Cloud

Underwater DCs (Project Natick), nuclear-powered Azure SMR, ocean/desert cooling; net-zero 2030 targets

🤖 Autonomous AI Ops

AI Agent-driven operations (AIOps), self-healing infrastructure, predictive capacity planning via ML

🔗 CXL Memory Pooling

CXL 3.0 enables disaggregated memory pools; breaks NUMA limits; shared memory across servers

Cloud Cost Optimization

47 FinOps — Engineering for Efficiency

30-50%
Typical TCO reduction vs. on-premises

80%
Potential savings with Spot Instances

40%
Wasted cloud spend (idle resources)

72%
Max savings with Reserved Instances

Right-Sizing

- Use AWS Compute Optimizer recommendations
- Downsize underutilized EC2 instances
- Match instance family to workload type (C-series=CPU, R-series=RAM, G-series=GPU)

Storage Lifecycle

- S3 Intelligent-Tiering moves objects automatically
- Transition to Glacier after 90 days (90% cost reduction)
- Delete unused EBS volumes and old snapshots

Network Cost

- Intra-AZ traffic: free. Cross-AZ: $0.01/GB
- Egress to internet: ~$0.09/GB (most expensive)
- Use CloudFront to reduce origin data transfer

FinOps Culture

- Tag all resources for cost allocation by team/env
- Set AWS Budgets alerts at 80% and 100%
- Weekly cost reviews; use Cost Explorer anomaly detection

AWS Well-Architected Framework

48 Six Pillars of Good Cloud Design

The AWS Well-Architected Framework provides a consistent approach for evaluating cloud architectures and implementing designs that scale over time. Six pillars guide every architectural decision.

⚙️ Operational Excellence

Run and monitor systems; continuously improve processes and procedures. IaC, runbooks, observability.

🔒 Security

Protect data, systems, assets. IAM least privilege, encryption, logging, automated incident response.

🔄 Reliability

Recover from failure, scale horizontally, manage change. Multi-AZ, backups, chaos engineering.

⚡ Performance Efficiency

Use compute resources efficiently. Right instance type, serverless where appropriate, benchmarking.

💰 Cost Optimization

Avoid unnecessary cost. Reserved/Spot pricing, right-sizing, lifecycle policies, FinOps culture.

🌱 Sustainability

Minimize environmental impact. Efficient utilization, renewable energy, managed services.

Course Summary — All Three Sections

49 Key Takeaways

Computer Organization

✓ Von Neumann: CPU + Memory + I/O + Bus; stored program concept

✓ ALU computes, CU orchestrates, Registers hold live data

✓ Memory hierarchy: Register → L3 Cache → RAM → SSD → HDD

✓ Pipelining overlaps stages; Amdahl's Law limits parallelism

Hardware Comparison

✓ Server CPUs: more cores, ECC, dual-socket vs. desktop single-socket

✓ RDIMM/LRDIMM forces ECC for 24/7 data integrity

✓ Data centers use 100G/400G networking and InfiniBand

✓ Tier IV: five-nines uptime, 2N power, immersion cooling PUE 1.02

Cloud Computing

✓ IaaS/PaaS/SaaS: trade control for convenience at each layer

✓ Type-1 Hypervisor + Docker containers enable multi-tenancy

✓ Auto Scaling + Load Balancer deliver elastic fault-tolerant apps

✓ Lambda serverless: pay per invocation, zero server management

AWS & Cloud Market

✓ AWS leads with 32% share, 200+ services, 33 global regions

✓ S3 (object) / EBS (block) / EFS (file) cover all storage patterns

✓ DevOps CI/CD: git push → test → build → deploy in minutes

✓ Security: Shared Responsibility, IAM, encryption, GuardDuty